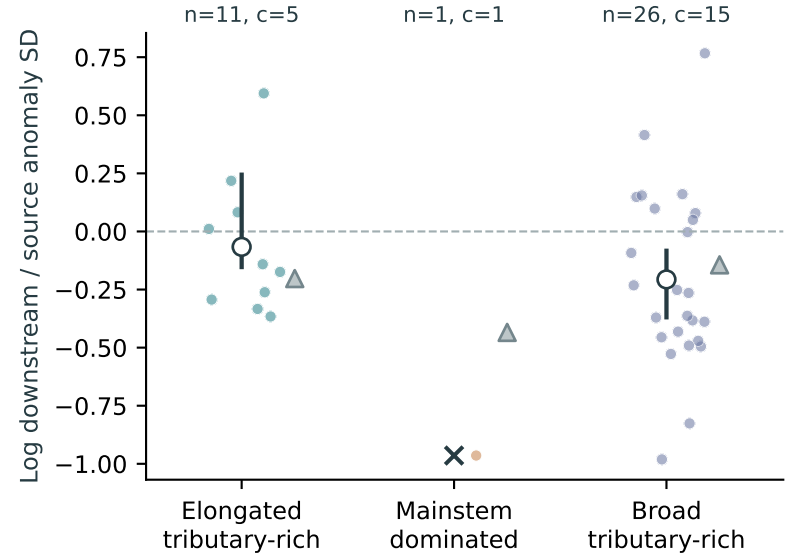


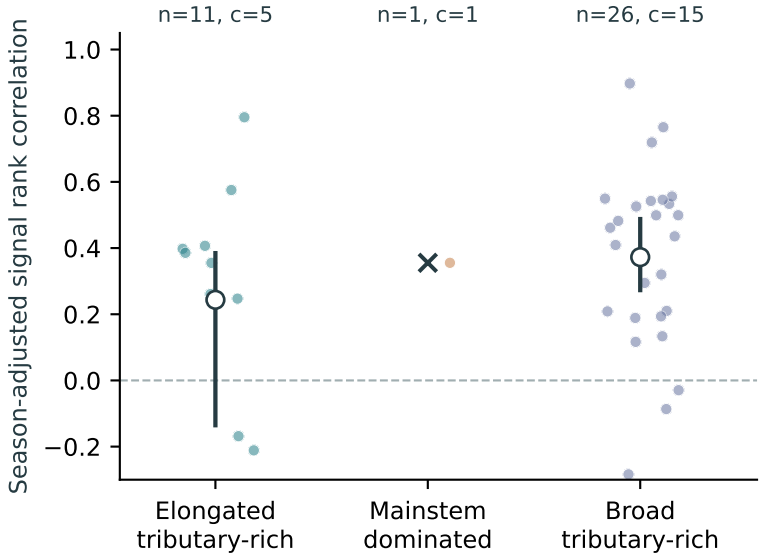
How do DOC fluctuations reach downstream stations in real river networks?

Panel a grey triangles: area-weighted two-source mixture signal

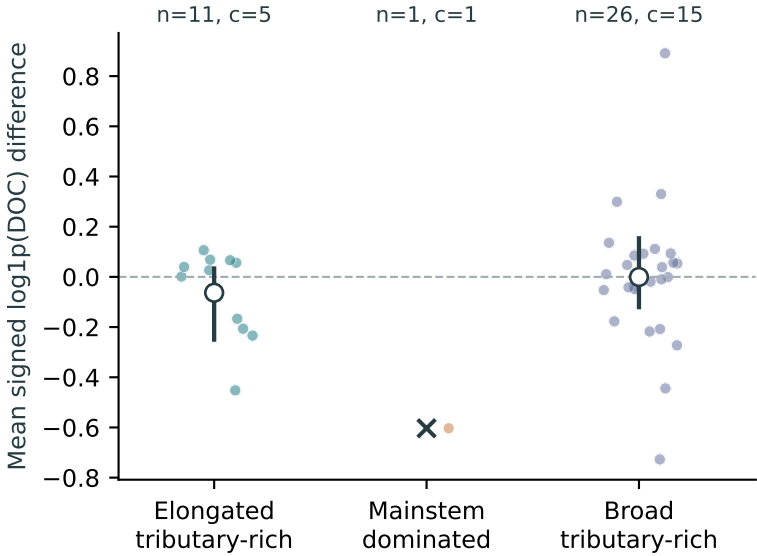
a Tributary integration



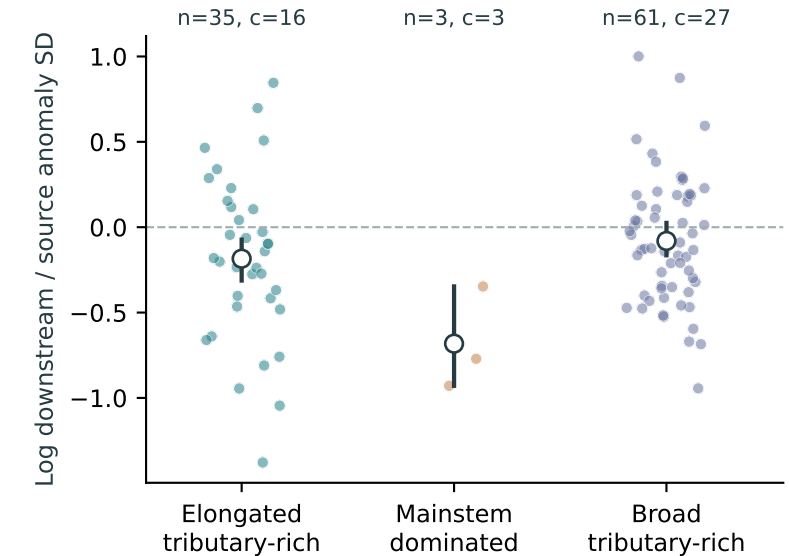
b Tributary integration



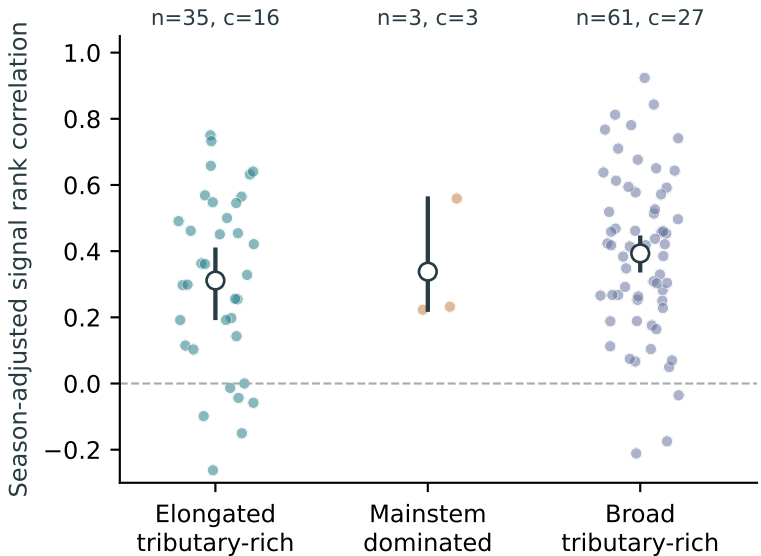
c Tributary integration



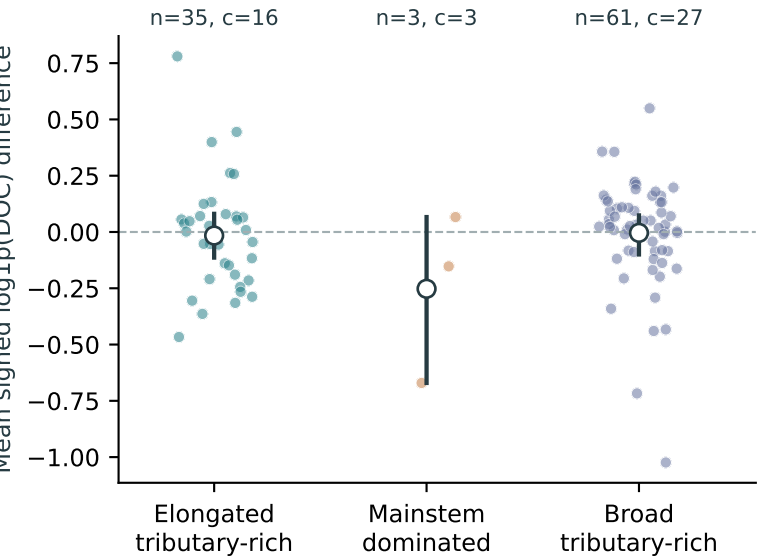
d Along-channel evolution



e Along-channel evolution



f Along-channel evolution



Dots: receiving stations. Open circles and intervals: connected-station bootstrap 95%; x: interval not estimable. n=receivers, c=systems. Monthly source-role log1p(DOC) anomalies, calendar/year adjusted. Negative SD ratios describe buffering, not removal. Classes are unadjusted.